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 So


```
In [1]: !pip install spacy
```

```
In [49]: import spacy
```

```
In [50]: nlp = spacy.load('en_core_web_sm')
```

```
In [11]: doc = nlp("I will google about facebook")
```

```
In [51]: doc.text
```

```
Out[51]: 'I will google about facebook'
```

```
In [54]: doc[-1]
```

```
Out[54]: facebook
```

```
In [14]: doc[0].pos1
```

```
Out[14]: 'PRON'
```

```
In [15]: doc[0].tag_
```

```
Out[15]: 'PRP'
```

In [23]: `spacy.explain('NNP')`

pos-tagging - Jupyter Notebook

localhost:8888/notebooks/pos-tagging.ipynb

pos-tagging Last Checkpoint: 40 minutes ago (autosaved)

FileEditViewInsertCellKernelWidgetsHelp

RunCode

TrustedPython 3

Logout

In [51]: doc.text

Out[51]: 'I will google about facebook'

In [54]: doc[-1]

Out[54]: facebook

In [57]: doc[2].pos_

Out[57]: 'VERB'

In [59]: doc[2].tag_

Out[59]: 'VB'

In [60]: spacy.explain('VB')

Out[60]: 'verb, base form'

In [24]: for word in doc:
print(word.text, "----->", word.pos_, word.tag_, spacy.explain(word.tag_))

I -----> PRON PRP pronoun, personal
will -----> AUX MD verb, modal auxiliary
google -----> VERB VB verb, base form
about -----> ADP IN conjunction, subordinating or preposition
facebook -----> PROPN NNP noun, proper singular

pos-tagging - Jupyter Notebook

pos-tagging.ipynb

pos-tagging Last Checkpoint: 41 minutes ago (unsaved changes)

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Logout

In [60]:

spacy.expain('verb, base form')

Out[60]:

'verb, base form'

In [61]:

```
for word in doc:
    print(word.text, "-----", word.pos_, word.tag_, spacy.explain(word.tag_))
```

Out[61]:

```
I -----> PRON PRP pronoun, personal
Will -----> AUX MD verb, modal auxiliary
google -----> VERB VB verb, base form
about -----> ADP IN conjunction, subordinating or preposition
facebook -----> PROPN NNP noun, proper singular
```

In [62]:

```
doc2 = nlp("I left the room")
for word in doc2:
    print(word.text, "-----", word.pos_, word.tag_, spacy.explain(word.tag_))
```

Out[62]:

```
I -----> PRON PRP pronoun, personal
left -----> VERB VBD verb, past tense
the -----> DET DT determiner
room -----> NOUN NN noun, singular or mass
```

In [29]:

```
doc3 = nlp("to the left of the room")
for word in doc3:
    print(word.text, "-----", word.pos_, word.tag_, spacy.explain(word.tag_))
```

Out[29]:

```
to -----> ADP IN conjunction, subordinating or preposition
the -----> DET DT determiner
left -----> NOUN NN noun, singular or mass
of -----> ADP IN conjunction, subordinating or preposition
```

pos-tagging - Jupyter Notebook

localhost:8888/notebooks/pos-tagging.ipynb

pos-tagging

Jupyter

pos-tagging

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Logout

RunCode

In [64]:

```
doc4 = nlp("I read books on history")
for word in doc4:
    print(word.text, "-----", word.pos_, word.tag_, spacy.explain(word.tag_))
```

```
I -----> PRON PRP pronoun, personal
read -----> VERB VBP verb, non-3rd person singular present
books -----> NOUN NNS noun, plural
on -----> ADP IN conjunction, subordinating or preposition
history -----> NOUN NN noun, singular or mass
```

In [65]:

```
docs = nlp("I have read a book on history")
for word in docs:
    print(word.text, "-----", word.pos_, word.tag_, spacy.explain(word.tag_))
```

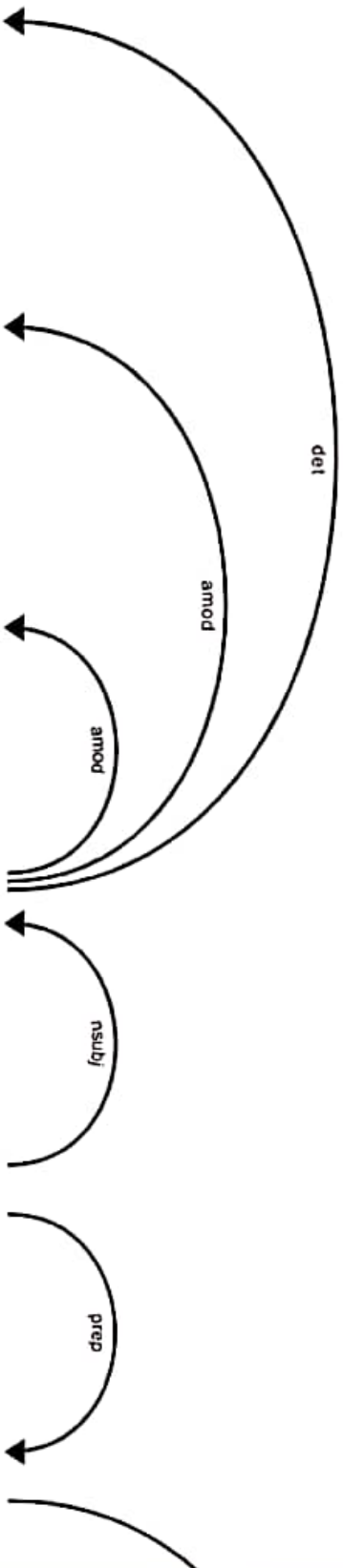
```
I -----> PRON PRP pronoun, personal
have -----> AUX VBP verb, non-3rd person singular present
read -----> VERB VBN verb, past participle
a -----> DET DT determiner
book -----> NOUN NN noun, singular or mass
on -----> ADP IN conjunction, subordinating or preposition
```

```
have -----> AUX VBP verb, non-3rd person singular present
read -----> VERB VBN verb, past participle
a -----> DET DT determiner
book -----> NOUN NN noun, singular or mass
on -----> ADP IN conjunction, subordinating or preposition
history -----> NOUN NN noun, singular or mass
```

```
In [66]: doc6 = nlp(u"The quick brown fox jumped over the lazy dog")
```

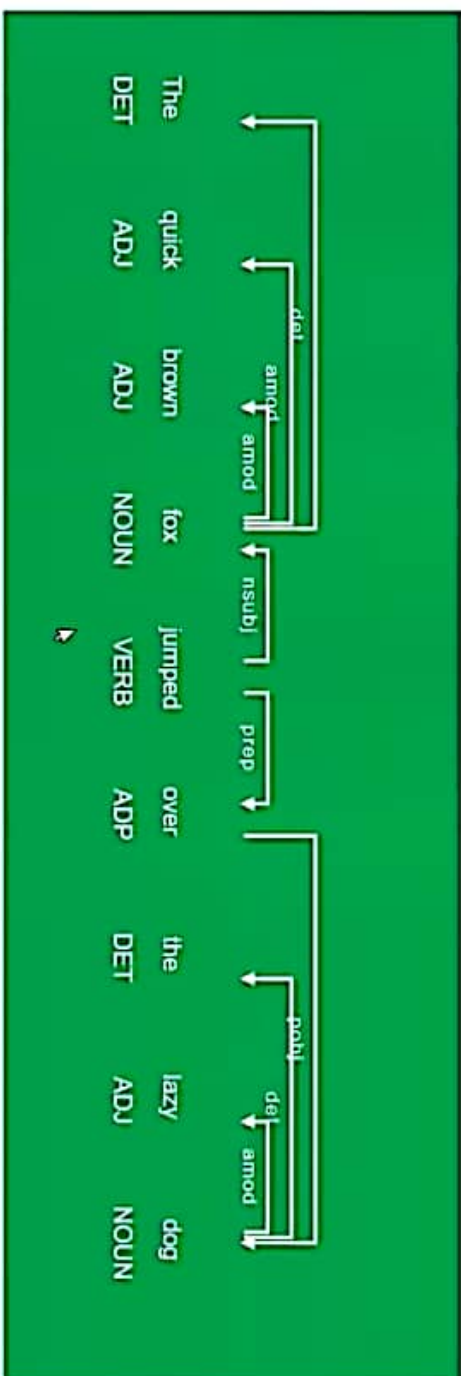
```
In [67]: from spacy import displacy
```

```
In [40]: displacy.render(doc6, style='dep', jupyter=True)
```



}

In [69]: `displacy.render(doc6, style='dep', jupyter=True, options=options)`



In []:

A friendly introduction to Bayes Theorem and Hidden Markov Models

Luis Serrano



A friendly introduction to Bayes Theorem and Hidden Markov Models
389,821 views · Mar 27, 2018

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SUMMARY

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0/20 Creator Suggested	(click to load) Ranked Tags	SEO Score
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510 Likes	1 Upvotes	(click to search) Mentions
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CHANNEL

4.58M Views	93K Subscribers	43 Videos
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☒ High Res. Thumbnail ☒ Info Cards Added ☒ End Screen Added	☒ Comment Pinned ☒ Liked on Facebook ☒ Captions Added	☒ Comment Hearted ☒ Chapters Added
---	---	---

TAGS

machine learning	probability	markov model	bayes theorem
hidden markov model	data science	mathematics	

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HMM →

[will will google campus]

N V M

{
 Nitish (N) loves (V) campus (N)
 can (M) Nitish (N) google (V) campus (N)
 will (M) Ankit (N) google (V) campus (N)
 Ankit (N) loves (V) will (N)

How POS Tagging works?

Thursday, January 20, 2022 1:50 PM

HMM →

[will will google campus]

N V M
Emission Prob

Nish (N) loves (V) campus (N)
 car (N) Nish (N) google (V) campus (N)
 will (M) Ankita (N) google (V) campus (N)
 Ankita (N) loves (V) will (N)
 will (N) loves (V) google (N)

	N	M	V
Nish	2	0	0
loves	0	0	3
campus	3	0	0
google	1	0	2
will	2	1	0
Ankita	2	0	0
car	0	1	0

How POS Tagging works?

Thursday, January 20, 2022 1:50 PM

HMN →

[will will google campus]

N V N
Emission Prob

Nitish (N) loves (V) campus (N)
 can (M) Nitish (N) google (V) campus (N)
 will (M) Ankita (N) google (V) campus (N)
 Ankita (N) loves (V) will (N)
 will (N) loves (V) google (N)

	N	M	V
Nitish	2/10	0	0
loves	0	0	3/5
campus	3/10	0	0
google	1/10	0	2/5
will	2/10	1/2	0
Ankita	2/10	0	0
can	0	1/2	0



How POS Tagging works?

Thursday, January 20, 2022 1:50 PM

HMM →

[will will google campus]



Emission

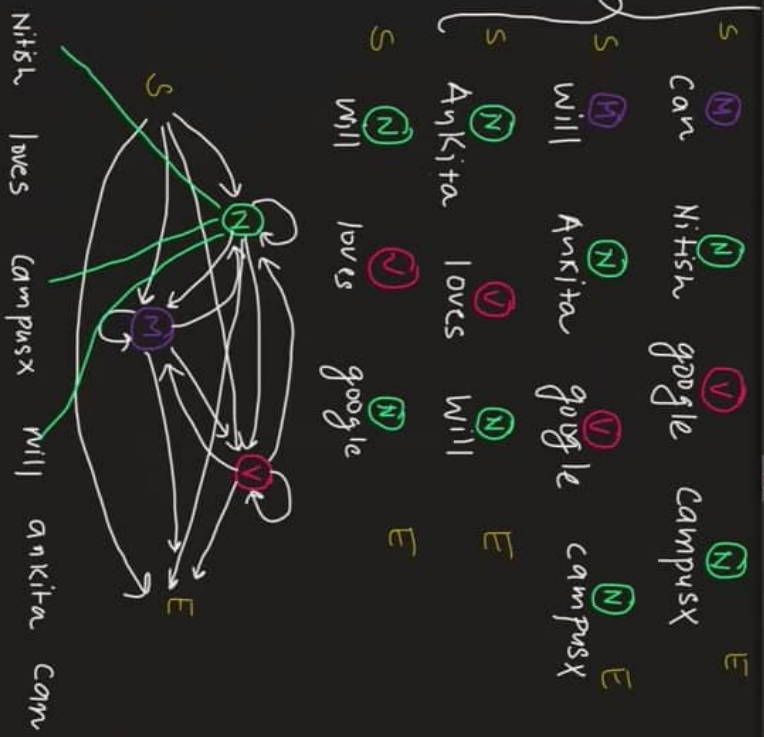
	N	M	V
Nitish	2/10	0	0
loves	0	0	3/5
campus	3/10	0	0
google	1/10	0	2/5
will	2/10	1/2	0
Aankita	2/10	0	0
can	0	1/2	0

Transition

	N	M	V	E
S	3/5	2/5	0	0
N	0	0	5/10	5/10
M	2/2	0	0	0
V	5/5	0	0	0

Emission Prob
Transition Prob





→ Emission

	N	M	V
Nishu	2/10	0	0
love	0	0	3/5
campus	3/10	0	2/5
google	1/10	0	0
will	2/10	1/2	0
Ankita	2/10	0	0
car	0	1/2	0

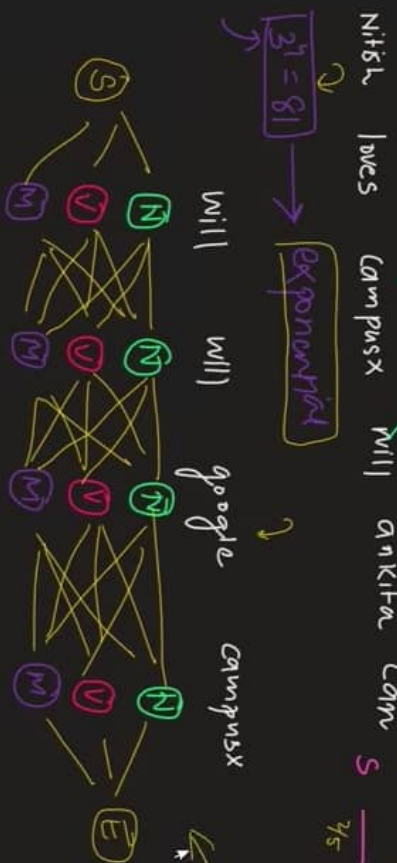
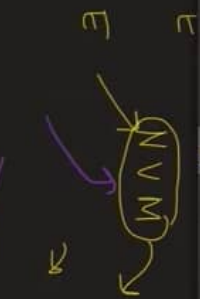
→ Transition

	N	M	V	E
S	3/5	2/5	0	0
N	0	0	5/10	5/10
M	2/2	0	0	0
V	5/5	0	0	0

will will google campus

$$\begin{aligned}
 & \frac{1}{5} \cdot \frac{1}{5} \cdot \frac{1}{10} \cdot \frac{3}{10} = 0 \\
 & \frac{3}{5} \cdot \frac{0}{5} \cdot \frac{1}{10} \cdot \frac{3}{10} = 0 \\
 & \frac{2}{5} \cdot \frac{0}{5} \cdot \frac{1}{10} \cdot \frac{3}{10} = 0 \\
 & \frac{2}{5} \cdot \frac{0}{5} \cdot \frac{1}{10} \cdot \frac{3}{10} = 0 \\
 & \frac{2}{5} \cdot \frac{0}{5} \cdot \frac{1}{10} \cdot \frac{3}{10} = 0 \\
 & \frac{2}{5} \cdot \frac{0}{5} \cdot \frac{1}{10} \cdot \frac{3}{10} = 0
 \end{aligned}$$



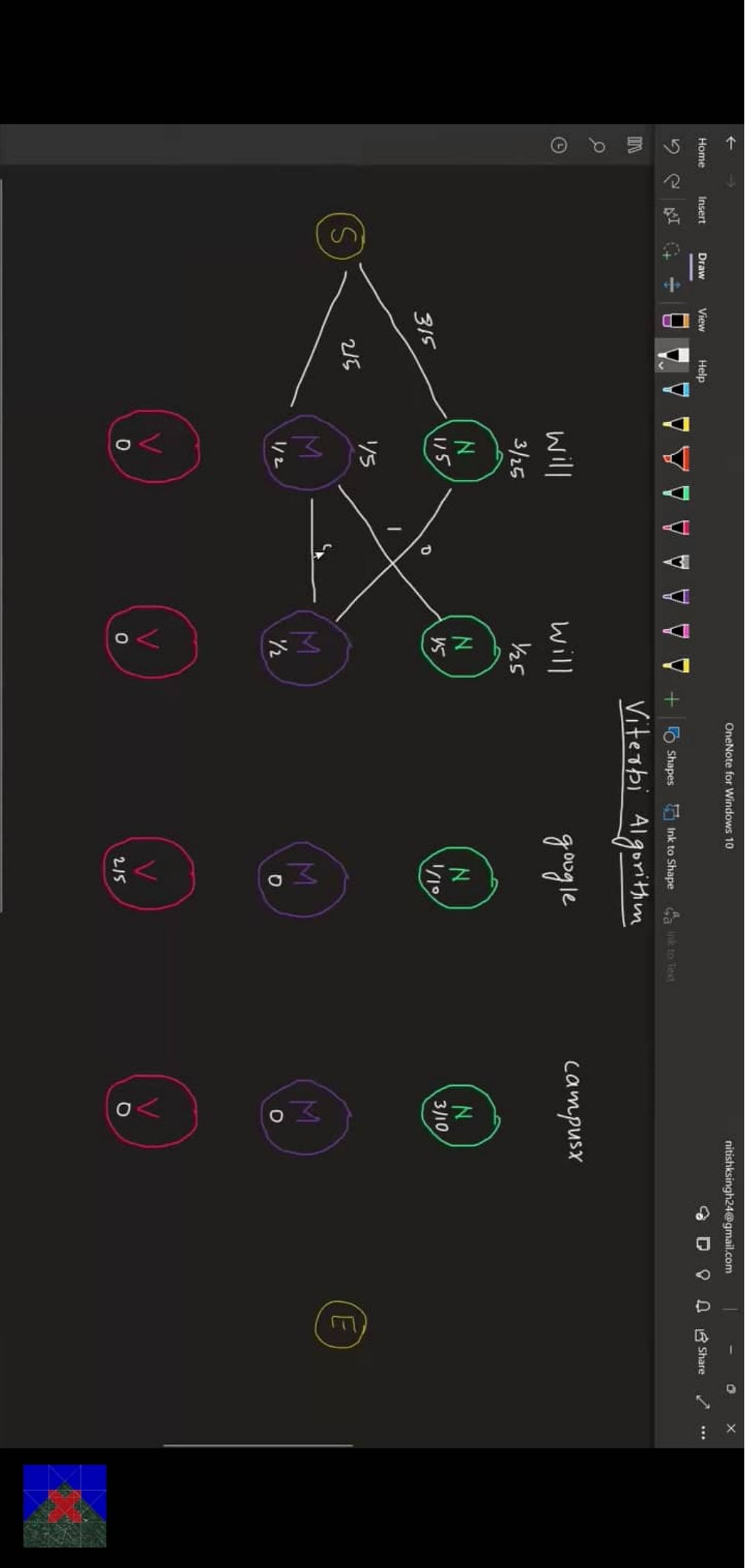


$$\begin{array}{c}
 \text{M} \\
 \text{will} \\
 \text{N} \\
 \text{will} \\
 \text{V} \\
 \text{google} \\
 \text{N} \\
 \text{campus's X}
 \end{array}$$

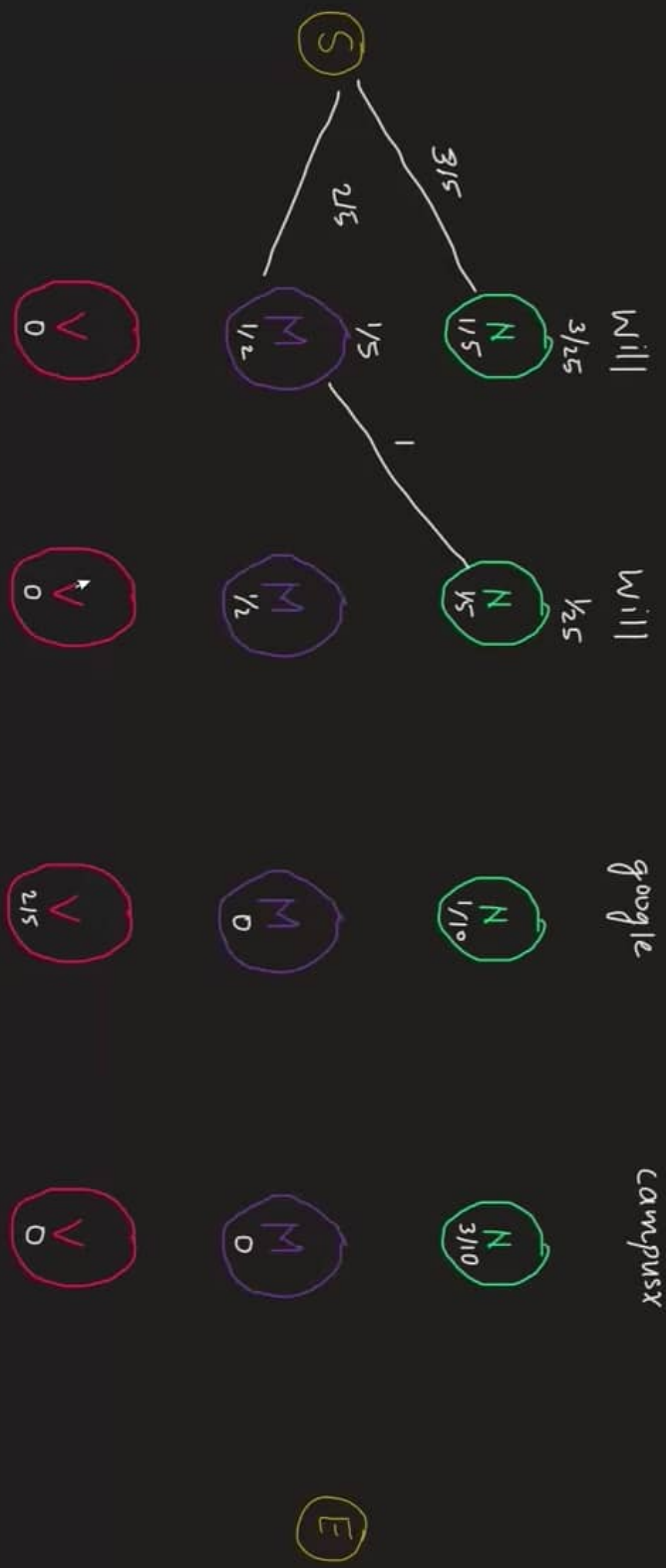
Optimization $\rightarrow$ Viterbi Alg^o



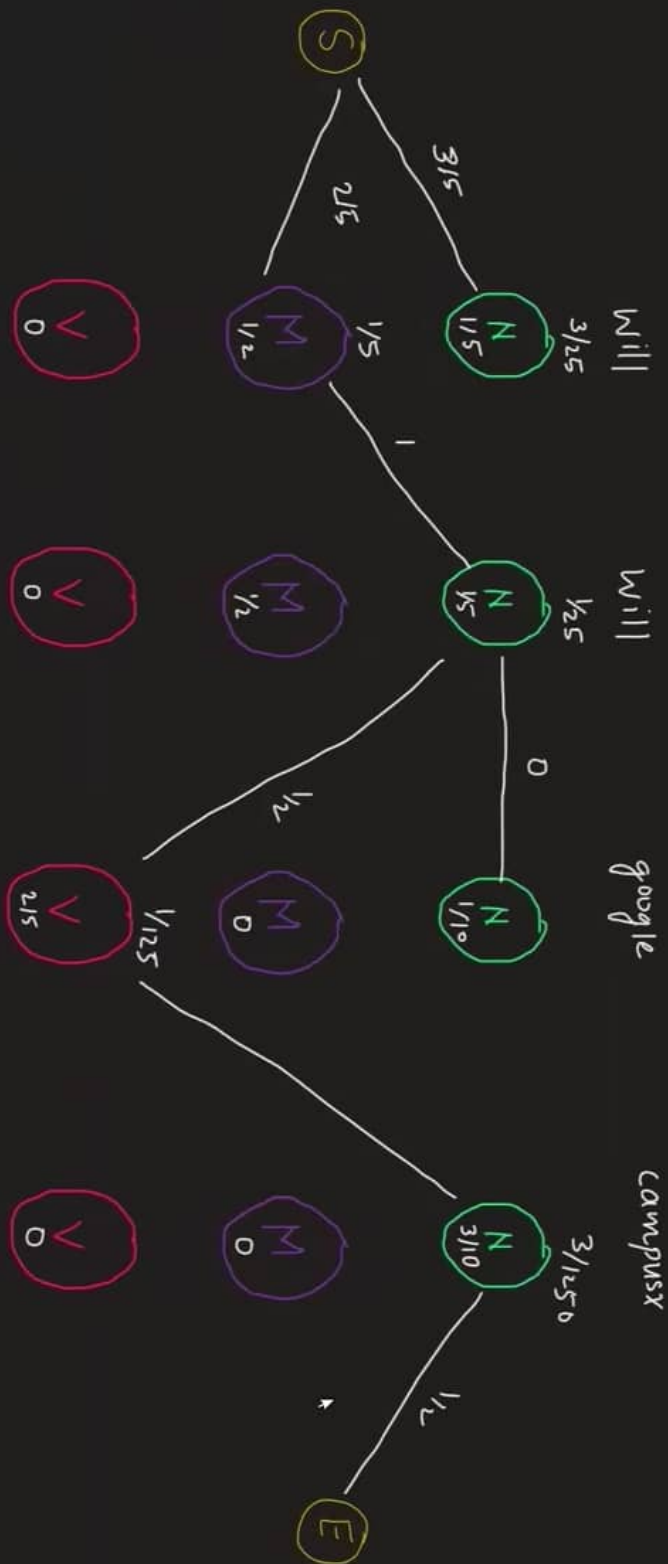


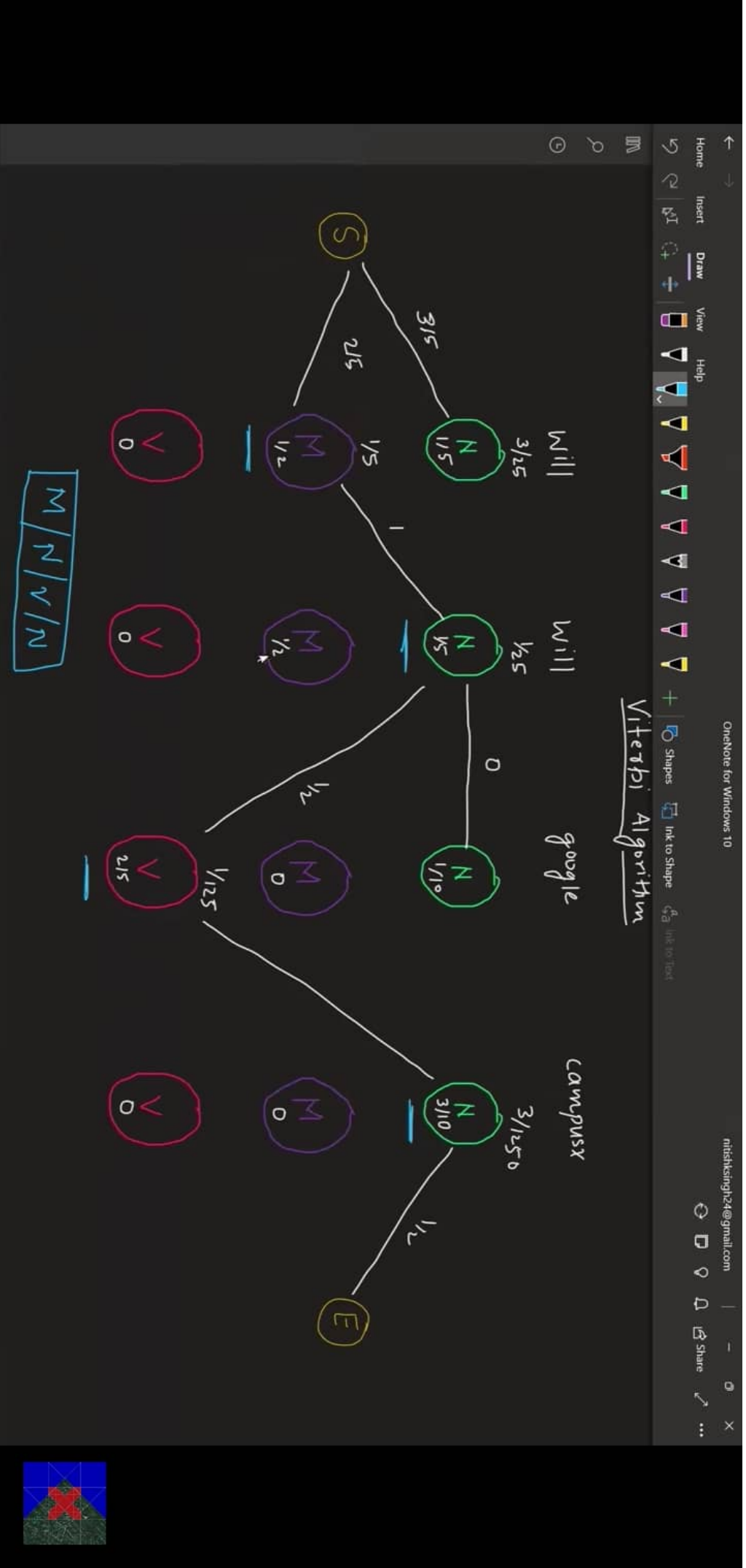


Viterbi Algorithm



Viterbi Algorithm







Home Page - Select or create a ...
9-Named Entity Recognition - ...
E-Prints Of Speech Tagging - ...
Text Preprocessing Singapore ...
6-Lemmatization- Test Prepro ...
5-Semantic And Its Types- Te ...
eiffel tower was built by whom ...

9-Named Entity Recognition (unsaved changes)
Logout

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Trusted | Python 3

Percent Eg: 20%, twenty percent
10
9
10

In [2]:
1 sentence="The Eiffel Tower was built from 1887 to 1889 by French engineer Gu
,

In [4]:
1 import nltk
2 words=nltk.word_tokenize(sentence)

In [6]:
1 tag_elements=nltk.pos_tag(words)

In []:
1 I

In []:
1

In []:
1

In []:
1

Type here to search



00:47
19-10-2022



Home Page - Select or create a ...
9-Named Entity Recognition - ...
Entity Of Speech Tagging - Au...
Text Preprocessing Script...
6-lemmatization- Test Prepro...
5-Semantic And its Types- Te...
entity tower was built by whom...
Logout

Jupyter 9-Named Entity Recognition (unsaved changes)

File Edit View Insert Cell Kernel Widgets Help
Trusted Python 3

In [4]:
1 import nltk
2 words=nltk.word_tokenize(sentence)

In [6]:
1 tag_elements=nltk.pos_tag(words)

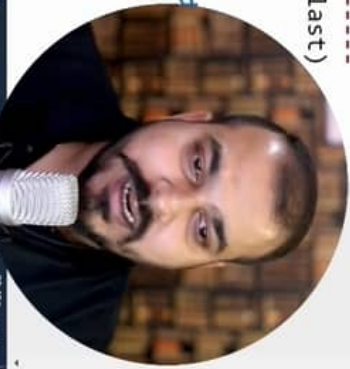
In []:
1 nltk.download('maxent_ne_chunker')

In [7]:
1 nltk.ne_chunk(tag_elements)

```
LookuperError
<ipython-input-7-99637582c7ec> in <module>
----> 1 nltk.ne_chunk(tag_elements)

Traceback (most recent call last)

~\anaconda3\lib\site-packages\nltk\chunk\_init_.py in ne_chunk(tagged
binary)
183     else:
184         chunker_pickle = _MULTICLASS_NE_CHUNKER
--> 185         chunker = load(chunker_pickle)
186         return chunker.parse(tagged_tokens)
```



```
In [8]: 1 nltk.download('maxent_ne_chunker')
```

```
[nltk_data] Downloading package maxent_ne_chunker to
[nltk_data] C:\Users\win10\AppData\Roaming\nltk_data...
[nltk_data] Unzipping chunkers\maxent_ne_chunker.zip.
```

Out[8]: True

```
In [7]: 1 nltk.ne_chunk(tag_elements)
```

LookUpError

```
<ipython-input-7-99637582c7ec> in <module>
```

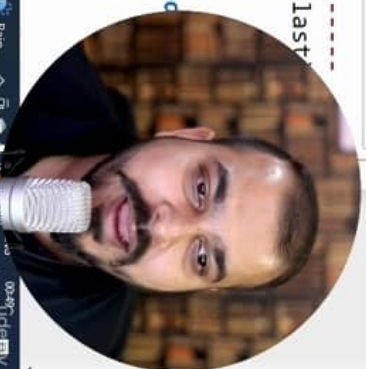
```
----> 1 nltk.ne_chunk(tag_elements)
```

```
~\anaconda3\lib\site-packages\nltk\chunk\__init__.py in ne_chunk(tagged
binary)
```

```
183 else:
```

```
chunker_pickle = MULTICLASS_NE_CHUNKER
```

```
--> 185 chunker = load(chunker_pickle)
```



Home Page - Select or create a... 9-Named Entity Recognition - 20Recognition.pynb

File Zoom

The DT ORGANIZATION was 100 null 100 100 Entity Name Tower 100

9-Named Entity Recognition (unsaved changes)

Logout

View Insert Cell Kernel Widgets Help Trusted Python 3

Run Code

```
In [10]: 1 nltk.download('words')
```

```
[nltk_data] Downloading package words to  
[nltk_data] C:\Users\win10\AppData\Local\Roaming\nltk_data...  
[nltk_data] Unzipping corpora\words.zip.
```

```
Out[10]: True
```

```
In [*]: 1 nltk.ne_chunk(tag_elements).draw()
```

```
In [ ]: 1
```

```
In [ ]: 1
```

```
In [ ]: 1
```

005377defely

19-10-2022 19:10